

**Biological Context** Wild-type enzyme, target function, structural data, known mutants



**Data Integration** Sequence/structure DBs, kinetic data, literature



→ **AI Reasoning Layer**

- Representation: PLMs, GNNs
- Reasoning: hypotheses, causality
- Prediction:  $\Delta\Delta G$ ,  $k_{\text{cat}}/K_m$



**In Silico Design** Ranked mutant list, multi-objective optimization



**Experimental Validation** Mutagenesis, enzyme assays, structural analysis

**Feedback & Learning** Update AI model, refine hypotheses, expand knowledge

